3212. Count Submatrices With Equal Frequency of X and Y

Given a 2D character matrix grid, where grid[i][j] is either 'X', 'Y', or '.', return the number of submatrices that contain:

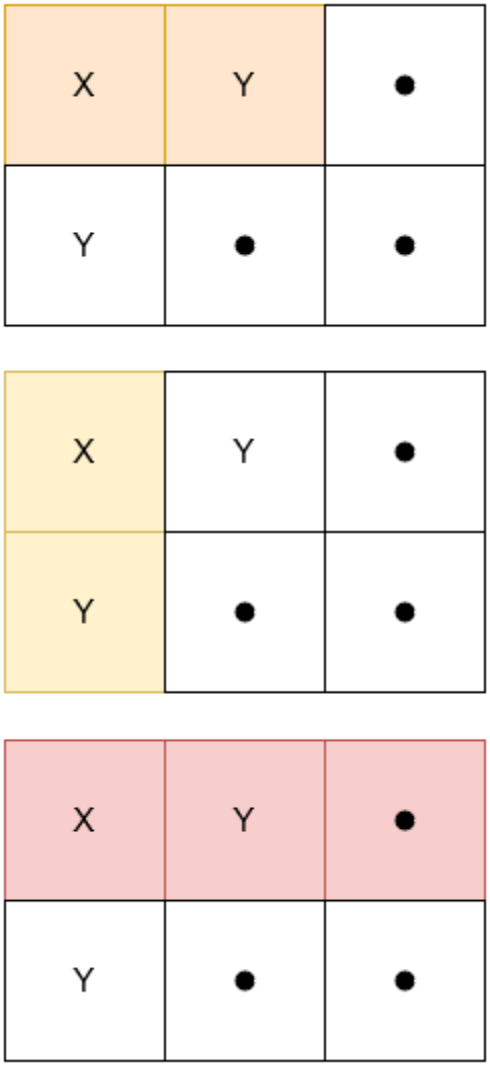
- grid[0][0]
- an **equal** frequency of 'X' and 'Y'.
- **at least** one 'X'.

Example 1:

Input: grid = [["X","Y",".."],["Y",".",".."]]

Output: 3

Explanation:



Example 2:

Input: grid = [["X","X"],["X","Y"]]

Output: 0

Explanation:

No submatrix has an equal frequency of 'X' and 'Y'.

Example 3:

Input: grid = [[".",".“],[".",".“]]

Output: 0

Explanation:

No submatrix has at least one 'X'.

Constraints:

- 1 <= grid.length, grid[i].length <= 1000
- grid[i][j] is either 'X', 'Y', or '.'.

Solution:--

```
class Solution {
public:
   int numberOfSubmatrices(vector<vector<char>>& grid) {
      int n = grid.size(), m = grid[0].size();
      int ans = 0;
      vector<vector<array<int, 2>>> sum(n + 1, vector<array<int, 2>>(m + 1, {0, 0}));

      for (int i = 0; i < n; i++) {
         for (int j = 0; j < m; j++) {

            int top = sum[i][j + 1][0];
            int left = sum[i + 1][j][0];
            int diag = sum[i][j][0];

            int val = left + top - diag;
            int hasY = sum[i][j + 1][1] | sum[i + 1][j][1];

            if (grid[i][j] == 'X') {
               val += 1;
               hasY = 1;
            }
            else if (grid[i][j] == 'Y') {
```

```
                val -= 1;
            }

            sum[i + 1][j + 1] = {val, hasY};

            if (val == 0 && hasY == 1) {
                ans++;
            }
        }
    }

    return ans;
  }
};
```